Lecture 9 – Software Quality Assurance

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1. Software Testing

Definition

Software testing is the process of executing a program with the intention of finding errors.

Key Points

- Goal is to find defects, not to prove correctness
- Testing is planned and systematic
- Largest consumer of SQA resources

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2. Testing vs Quality Assurance (QA)

Testing

- Subset of QA
- Product oriented
- Reactive activity
- Focuses on identifying defects

Quality Assurance

- Broader than testing
- Process oriented
- Proactive activity
- Focuses on preventing defects

Exam Line: Testing finds bugs, QA prevents bugs.

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3. Defects

Definition

A defect is any flaw or issue in software that causes it to deviate from expected behavior or requirements.

Notes

- Also called bugs or errors
- Discovered through testing
- Can be functional or non functional

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4. Manual Testing vs Automation Testing

Manual Testing

- Executed by humans
- No scripts or tools
- Flexible and exploratory

Advantages

- Easy to adapt to requirement changes
- Good for UI and exploratory testing

Disadvantages

- Human error
- Time consuming
- Expensive for large projects

Automation Testing

- Executed using tools and scripts
- Faster and repeatable

Advantages

- Speed
- Reusability

- Immediate results

Disadvantages

- Initial setup time
- Maintenance overhead
- Not suitable for frequently changing requirements

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5. When to Use Manual vs Automation

Situation	Preferred Approach
Exploratory testing	Manual
Regression testing	Automation
One time testing	Manual
Repetitive testing	Automation

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6. Baselines

Definition

A baseline is an approved reference point used to compare future changes.

Purpose

- Maintain consistency
- Control changes
- Track progress

Types of Baselines

1. Requirements Baseline
    - Approved requirements document

2. Design Baseline
    - Approved architecture and UI designs
3. Code Baseline
    - Stable version of code in version control
4. Test Baseline
    - Approved test cases and test plans
5. Product Baseline
    - Final released product

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7. Test Cases

Definition

A test case is a set of conditions or variables under which a tester determines whether a system works correctly.

Components of a Test Case

- Test Case ID
- Description
- Test Steps
- Test Data
- Expected Result
- Actual Result
- Pass or Fail Status

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8. Test Suites

Definition

A test suite is a collection of test cases grouped together to test a specific functionality or feature.

Example

Login test suite may include:

- Valid login
- Invalid login
- Empty fields

- Logout

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9. Test Coverage

Definition

Test coverage measures how much of the software has been tested.

Notes

- Indicates extent of testing
- High coverage does not guarantee bug free software
- Low coverage increases risk

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10. Requirement Coverage

Definition

Requirement coverage ensures that every requirement is tested by at least one test case.

Importance

- Ensures no requirement is missed
- Improves confidence in testing

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11. Requirement Traceability Matrix (RTM)

Definition

RTM is a table that maps requirements to test cases.

Purpose

- Identify uncovered requirements
- Track testing progress

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12. Test Driven Development (TDD)

Definition

Test Driven Development is a development approach where test cases are written before the actual code.

Steps

1. Write test case
2. Run test (fails)
3. Write code
4. Test passes
5. Refactor code

Benefits

- Early defect detection
- Better design
- High test coverage

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13. White Box vs Black Box Testing

Black Box Testing

- Internal code not visible
- Focus on inputs and outputs
- Based on requirements

White Box Testing

- Internal code visible
- Focus on logic and structure
- Coverage based testing

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14. Types of Testing (High Level)

- Unit Testing
- Integration Testing

- System Testing
- Acceptance Testing
